Meta-Adversarial Despeckling Network for Attacking SAR Image Target Detectors

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***Abstract*— Recently, integrated deep learning methods cascading speckle filtering and target detection have garnered increasing attention in synthetic aperture radar (SAR) image target detection. These methods use despeckling networks to suppress speckle noise, enhancing the performance of followed detectors. However, recent studies have shown that deep neural networks (DNNs) are vulnerable to adversarial attacks, where adding imperceptible perturbations to benign examples can cause incorrect predictions. This phenomenon raises serious concerns about the security of current integrated deep learning algorithms in SAR images. To this end, this work conducts adversarial attack research on integrated speckle filtering and detection methods to assess its adversarial robustness. Specifically, a novel meta-adversarial despeckling network (Meta-ADNet) architecture is proposed, which leverages the despeckling network as a potential attack pathway to inject perturbations, generating adversarial despeckled examples that invalidate subsequent detection processes. Meta-ADNet consists of two core components: the baseline model adversarial despeckling network (ADNet) and the corresponding meta-adversarial attack framework. ADNet first generates benign despeckled SAR images through a despeckling network, which are then fed into the perturbation injection branch. This branch guides the despeckling network in producing adversarial examples by backpropagating the confidence loss of the surrogate detection model. The meta-adversarial attack framework constructs different tasks by selecting multiple surrogate detection models and iteratively simulates white-box ensemble attacks and black-box attacks within each task to enhance the transfer attack capability of ADNet on black-box models. Extensive experimental results on the SAR detection datasets SSDD and HRSID demonstrate that the proposed algorithm can effectively attack white-box surrogate detectors and exhibits strong black-box transfer attack capabilities.**

***Index Terms*— Adversarial attacks, deep learning, metalearning, synthetic aperture radar (SAR), speckle filter, target detection.**

I. Introduction

SYNTHETIC aperture radar (SAR), as an active microwave sensor, possesses the capability for all-day and all-weather observation and imaging. It has been widely utilized in civilian and military domains, with numerous successful applications including natural disaster evaluation, land cover mapping, target detection, and recognition [1], [2], [3], [4], [5]. In recent years, there has been rapid development in radar imaging technology [6], leading to a significant improvement in the resolution of SAR images. As a result, there is increasing attention to efficiently and accurately detecting targets from massive high-resolution SAR images.

Traditional SAR image target detection algorithms mostly rely on the statistical characteristics of pixels in SAR images [7], [8], [9], [10], [11]. They distinguish between targets and backgrounds by adaptively setting thresholds based on statistical models of background clutter distributions. For instance, Rayleigh distribution [10] and G^0 distribution [11] are commonly used statistical models for clutters. However, limited by modeling accuracy and tunable parameters, these methods suffer from many false alarms and misses in SAR images with complex backgrounds.

Recently, deep neural networks (DNNs) have garnered increasing attention and achieved outstanding performance in target detection tasks due to their powerful feature representation and learning capabilities [12], [13], [14], [15], [16], [17], [18], [19], [20]. Drawing inspiration from target detection algorithms in optical images, DNN-based target detection algorithms in SAR images can also be categorized into two- and one-stage methods [21], [22], [23], [24], [25]. Two-stage methods [21], [22] first generate numerous category-agnostic region proposals to roughly localize targets, followed by classification and regression on candidate target regions. In contrast, one-stage methods [23], [24], [25] skip the region proposal generation step and transform the target bounding box localization problem into a regression problem, thereby achieving higher detection efficiency. Although the above methods have made significant progress, the inherent speckle noise [26], [27], [28] in SAR images limits their detection accuracy in complex interference backgrounds [28]. To address this issue, researchers have developed integrated approaches combining despeckling and detection, aiming to suppress speckle noise interference in SAR images by performing speckle filtering either before or concurrently with target detection [28], [29], [30], [31], [32]. These methods achieved better detection performance while obtaining higher visual quality SAR images.

However, recent research [33], [34], [35] indicates that DNN-based object detectors are vulnerable to adversarial examples: adding imperceptible perturbations to benign samples can render the detector ineffective. In optical images, researchers typically calculate classification and regression

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losses for candidate target regions based on the model's architecture and parameters. Adversarial samples are then iteratively generated by following the gradient ascent direction. Methods such as dense adversary generation (DAG [35]), robust adversarial perturbation (RAP [36]), and TOG [37] exemplify this approach. Unlike these methods, adversarial attacks on SAR images are predominantly focused on target recognition tasks. Given the unique characteristics and imaging mechanisms of SAR images, there has been limited research on adversarial attacks specifically targeting object detection in this domain [38], [39]. In particular, Zhou et al. [38] investigated the performance impact of adversarial perturbations generated by adversarial recognition algorithms [34] on target detectors. The work of Zhou et al. [39] proposed a conditional random field-based adversarial attack method for the interaction between targets and background in SAR images, which iteratively optimizes the perturbation by maximizing the energy difference between neighboring pixels. To the best of our knowledge, there is no research on adversarial attacks against the cascade despeckling and target detection framework.

In this work, we explore the impact of despeckled images as implicit adversarial examples on the performance of subsequent cascade detection networks and propose a novel meta-adversarial despeckling network (Meta-ADNet) framework. Meta-ADNet comprises two core components: the baseline network ADNet and the meta-adversarial attack framework, aiming to equip despeckled images with adversarial attack capabilities and further enhance their cross-model transferability. As for ADNet, it achieves the goal of outputting adversarial despeckled examples by modifying the loss function of the despeckling network. Specifically, a confidence loss is introduced to corrupt the process of distinguishing the target from the background by the surrogate detector. Through backpropagation, the despeckling network is guided to produce adversarial despeckled examples where the target confidence of the detector is below a given threshold, leading to the targets being discarded. The example of an adversarial attack on YOLOv3 [13] is shown in Fig. 1, where it can be seen that only a partial perturbation needs to be added to the despeckled image to cause the detector to lose most of the targets.

In addition, considering the limited transferability of adversarial examples to unseen black-box models [40], [41], [42], [43], [44], inspired by the metagradient adversarial attack algorithms [42], the proposed Meta-ADNet further incorporates meta-adversarial attack framework on top of ADNet. Meta-ADNet randomly samples multiple surrogate detection models from the model set to constitute multiple tasks, each consisting of two phases: metatraining and metatesting. During the metatraining phase, the confidence losses of the sampled surrogate models are combined using an ensemble approach to train the despeckling network to generate adversarial samples. In the metatesting phase, a black-box attack scenario is simulated using the preliminarily trained despeckling network to attack an unseen surrogate model. The despeckling network is then further optimized through gradient feedback to produce adversarial samples with stronger transferability. Notably, our method is trained end-to-end, eliminating the need for additional iterative optimization time and efficiently producing adversarial samples. The main contributions of this article are summarized as follows.

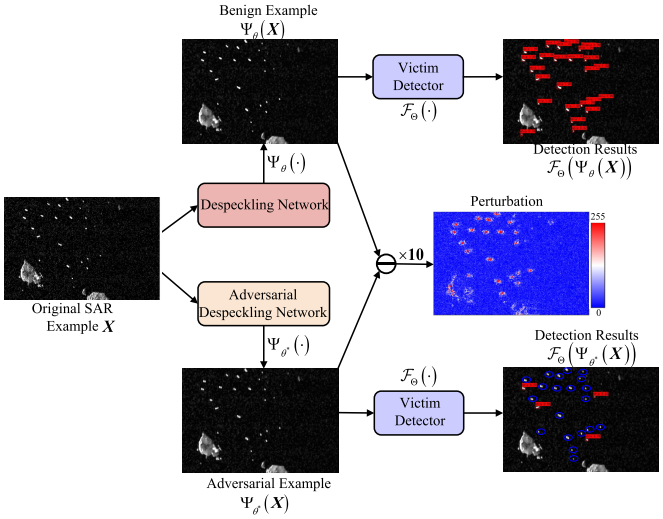


Fig. 1. Example of adversarial attack against the YOLOv3 detector, which is generated directly by the adversarial despeckling network. The perturbations are amplified tenfold for visual clarity. Red boxes indicate correct detections, yellow ellipse boxes indicate false positives, and blue ellipse boxes indicate false negatives.

1) We introduce a novel approach to attacking SAR image target detection networks using despeckled images and design a baseline adversarial despeckling network (ADNet).
2) Inspired by metalearning principles, a new framework for the Meta-ADNet is further proposed to enhance the transferability of adversarial despeckled examples by iteratively simulating white- and black-box attacks.
3) Extensive experimental results demonstrate that the proposed method exhibits strong cross-model transfer attack capabilities while significantly improving the overall perceptual quality of SAR images.

II. Related Work

A. Deep Learning for Object Detection in SAR Images

In recent years, with the expansion of SAR target detection datasets, deep learning-based target detection algorithms have rapidly developed and achieved significant success, evolving into numerous custom detectors tailored for different targets. Some studies have improved the network's multiscale detection performance by constructing feature pyramids. For instance, Zou et al. [45] utilized subaperture decomposition to mine multiscale features of images, achieving high-precision vehicle target detection by balancing multilayer semantic information. Zhao et al. [46] proposed an attention feature refinement and alignment network, which enhances the performance of SAR image aircraft target detection by aligning refined multiscale features. Bai et al. [24] used deformable convolution to align and fuse feature layers of different scales, enhancing the contour information of ship targets. Considering the sparse distribution of targets in SAR images, which leads to a large number of redundant anchor boxes, anchor-free

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detection methods have been introduced into SAR target detection. Among these, Ma et al. [47] proposed an anchor-free detection method based on keypoint estimation and attention mechanisms to improve the localization accuracy of ship targets. Kang et al. [48] added a scattering point prediction branch to the classification branch to capture the strong scattering points of aircraft targets and enhance the scattering topology structure of the aircraft targets by computing the spatial correlation between the scattering points. Recently, researchers have noted that the inherent speckle noise [26] in SAR images is a potential factor that degrades the performance of detection algorithms and proposed numerous solutions [28], [29], [30], [49], [50], [51]. Zhou et al. [49] constructed a feature extraction layer by combining average and max pooling to reduce the impact of strong scattering points on ships. Zhou et al. [50] have introduced a semantic segmentation branch to incorporate the edge semantic information of ships into the training process, guiding the model to focus more on ship targets. Zhang et al. [51] suppressed background noise by leveraging frequency domain information of SAR images. In addition to the methods above, speckle filtering [52], [53], [54], [55], [56], [57] has also received widespread attention either before or simultaneously with detection tasks. For example, Zhao et al. [28] integrated a speckle filtering branch into a rotated bounding box detection network to learn jointly denoising and detecting targets in SAR images in a multitask learning paradigm. Long et al. [30] designed a speckle reduction branch to assist the detection network in extracting clean features. Choi et al. [29] proposed a noise-robust cascaded framework, where the filtering network outputs denoised images conducive to ship detection.

We note that current DNN-based target detection methods primarily focus on improving the detection accuracy and operational efficiency of models, with little attention given to the security and vulnerability of the models. For instance, in the aforementioned despeckling cascaded detection algorithms, injecting adversarial perturbations during the despeckling process poses a potential security risk.

B. Adversarial Attack

Szegedy et al. [33] first revealed the vulnerability of CNN models in image classification tasks by introducing the concept of adversarial examples. Subsequently, researchers have developed numerous adversarial attack methods to deceive image classifiers. Depending on the attacker's knowledge of the target model, adversarial attacks can be categorized into white- and black-box attacks. White-box attacks occur when the attacker has full knowledge of the target model's structure, gradients, and training data, using this information to generate adversarial samples. For instance, methods such as FGSM [34] utilize gradient information to generate adversarial samples through backpropagation, while methods such as MI-FGSM [40] and project gradient descent (PGD [58]) enhance the attack effectiveness by optimizing the gradient update direction. On the other hand, black-box attacks refer to situations where the attacker has no access to the target model's architecture or parameters. In this case, the attacker can only interact with the model to obtain its predictions or generate adversarial samples by training a surrogate model to transfer the attack to the target model. Representative methods include feature disruptive attack (FDA [43]) and Meta-Attack [44].

Unlike image classification tasks, conducting adversarial attacks on object detection is more challenging due to the complex model mechanisms involved. Xie et al. [35] were the first to introduce adversarial examples into object detection tasks, proposing the DAG algorithm. This algorithm assigns an incorrect label to each proposal and generates adversarial examples through iterative gradient backpropagation. Li et al. [36] proposed RAP that disrupts the RPN by combining localization regression loss and classification loss, thus attacking two-stage detectors. Chow et al. [37] introduced the adversarial objectness gradient attack method, presenting three attack modes: object vanishing, object fabrication, and object mislabeling. Cai et al. [59] considered the co-occurrence and relationship among multiple objects and proposed a context-aware attack method to improve the success rate of transfer attacks. However, these methods rely on gradient backpropagation and add perturbations to the entire image, leading to high computational costs. To address this issue, Wei et al. [60] proposed a universal efficient adversary (UAE) that uses generative adversarial networks to quickly generate adversarial perturbations. Chen et al. [61] introduced a Shapeshifter that optimizes perturbations within patches to deceive detectors.

In contrast to the optical domain, an adversarial attack on SAR images has just begun [38], [39]. Designing attack algorithms that account for the unique characteristics of SAR images presents a significant challenge. In this article, we leverage the current deep learning-based SAR image despeckling networks and propose a novel adversarial despeckling attack detection network method.

C. Metalearning

Metalearning is often understood as learning to learn. It trains models by constructing different tasks, enabling them to acquire prior knowledge. When encountering new tasks, with only a small amount of data fine-tuning, the model can better generalize to new domains and quickly complete the task [62]. As a powerful learning approach, metalearning has been widely applied to various tasks [63], [64], including adversarial attack tasks. For instance, Du et al. [44] proposed a Meta-Attack, which uses a metatraining framework to train a gradient estimation network. When attacking unknown black-box models, it can deceive image classifiers with only a few queries. MetaAdvDet constructs multiple metatasks using adversarial examples generated by different attack algorithms to train a robust adversarial example detection network. Yuan et al. [42] proposed a metagradient adversarial attack (MGAA) framework that constructs different tasks by sampling multiple models from a model set. Within each task, it iteratively simulates black- and white-box attacks to generate adversarial examples.

Unlike MGAA [42], which focuses on attacking image classifiers, our proposed Meta-ADNet is the first

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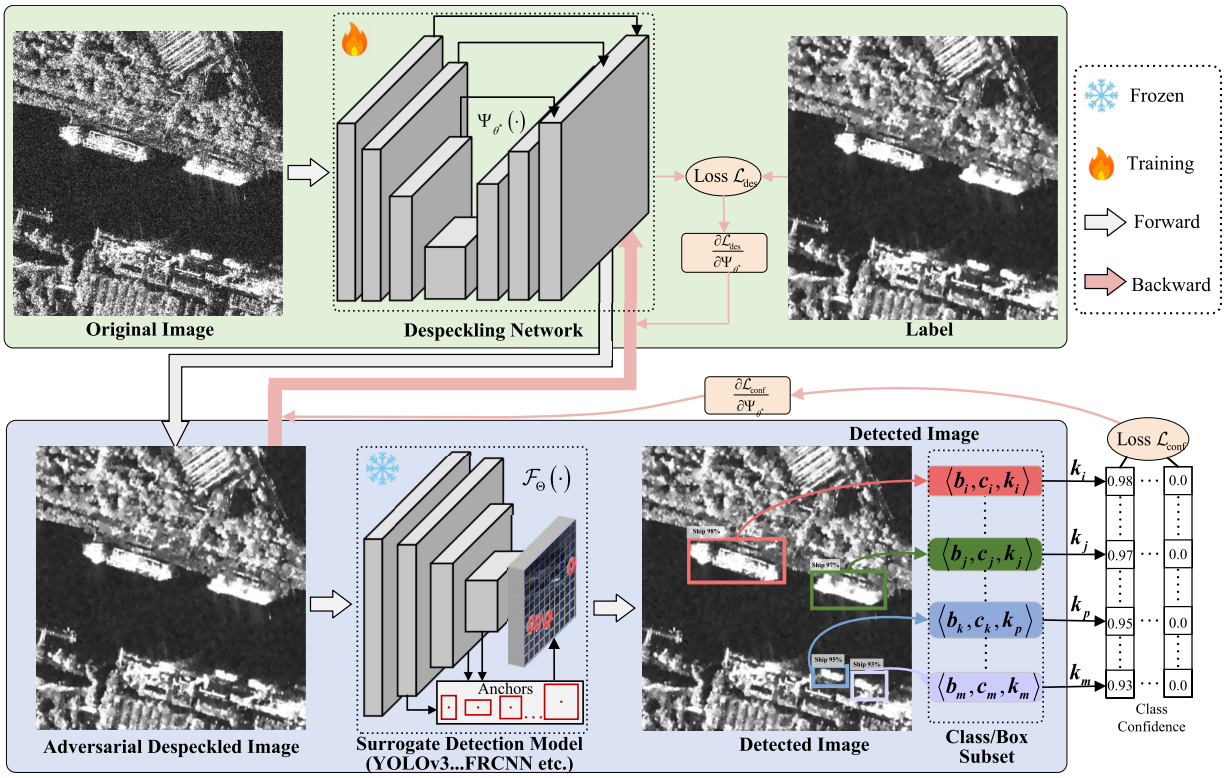


Fig. 2. Training framework of ADNet. ADNet uses the despeckling loss $\mathcal{L}_{\text{des}}$ (comparing the original SAR images with the clean labels, light green area) and confidence loss $\mathcal{L}_{\text{conf}}$ (reducing the confidence of the surrogate detection model, light blue area) to jointly optimize the despeckling network. During the testing phase, the despeckling network directly generates adversarial despeckled images to attack detectors with different architectures.

metalearning-based algorithm to conduct adversarial attacks on object detection. In addition, the target detection models sampled in the metatasks are used to train the despeckling network rather than iterating multiple times on the same sample internally, resulting in higher efficiency in generating adversarial despeckling images.

III. Methodology

A. SAR Speckle Modeling and Problem Statement

1) SAR Speckle Modeling: The multiplicative speckle in SAR images is the primary cause of degrading image quality. For an original SAR intensity image $\mathbf{X} \in \mathbb{R}^{H\times W}$ with a size of $H \times W$, according to the fully developed speckle model proposed by Goodman [26], $\mathbf{X}$ can be expressed as

$$\mathbf{X} = \mathbf{Y} \odot \mathbf{Q} \tag{1}$$

where $\odot$ denotes the Hadamard product, $\mathbf{Y}$ is the potential clean (despeckled) image, and $\mathbf{Q}$ represents speckle, assumed to follow a Gamma distribution with a unit mean and variance of $1/L$. The probability density distribution of $\mathbf{Q}$ is given by

$$p(\mathbf{Q}) = \frac{L^L \mathbf{Q}^{L-1} e^{-L\mathbf{Q}}}{\Gamma(L)} \tag{2}$$

where Γ represents the gamma distribution and $L \geq 1$ denotes the number of looks in the SAR imaging process, with smaller L values indicating stronger speckle. The goal of despeckling is to infer the despeckled image $\mathbf{Y}$ based on the original observed image $\mathbf{X}$, providing technical support for subsequent research on downstream tasks such as target detection.

2) Problem Statement of Adversarial Despeckling: Given an original SAR image $\mathbf{X} \in \mathbb{R}^{H\times W}$, despeckling networks $\Psi_\theta(\cdot)$, and adversarial despeckling network $\Psi_{\theta^*}(\cdot)$ with parameters θ and θ^*, respectively, then the despeckled image and the adversarial despeckled image can be denoted as $\Psi_\theta(\mathbf{X})$ and $\Psi_{\theta^*}(\mathbf{X})$. We hope that $\Psi_{\theta^*}(\mathrm{X})$ could attack the target detector $\mathcal{F}_\Theta(\cdot)$ with parameter Θ. Specifically, set $\Psi_\theta(\mathbf{X})$ to contain N ground-truth bounding boxes $\mathcal{O} = \{\boldsymbol{o}_i, \boldsymbol{c}_i\}_{i=1}^N$, where $\boldsymbol{o}_i = (b_i^x, b_i^y, b_i^h, b_i^w)$, and b_i^x, b_i^y and b_i^h, b_i^w denote the center coordinates of the ith bounding box and its corresponding length and width, respectively. $\boldsymbol{c}_i = \{c_i^1, c_i^2, \ldots, c_i^z\}$ and $c_i^m = 1$ if $\boldsymbol{o}_i$ is a class object m. For a ground-truth target $(\boldsymbol{o}_i, \boldsymbol{c}_i)$, assume that $\mathcal{F}_\Theta(\cdot)$ successfully detects the target on $\Psi_\theta(\mathbf{X})$ with $\mathrm{IoU}(\boldsymbol{o}_i, \boldsymbol{o}_i^{\Psi_\theta}) > 0.5$ and $\boldsymbol{c}_i = \boldsymbol{c}_i^{\Psi_\theta}$. For the adversarial sample $\Psi_{\theta^*}(\mathbf{X})$, whose detection result is $(\boldsymbol{o}_i^{\Psi_{\theta^*}}, \boldsymbol{c}_i^{\Psi_{\theta^*}})$ (which can be empty; $\boldsymbol{o}_i^{\Psi_{\theta^*}} = \emptyset$), if it satisfies that the IoU between $\boldsymbol{o}_i^{\Psi_{\theta^*}}$ and $\boldsymbol{o}_i$ is less than 0.5 or $\boldsymbol{c}_i \neq \boldsymbol{c}_i^{\Psi_{\theta^*}}$, this means a successful attack on the detector.

B. Adversarial Despeckling Network

In this section, we will first introduce the overall framework of the proposed baseline model ADNet, followed by a detailed explanation of the loss functions.

The training pipeline of ADNet is schematically shown in Fig. 2. Generally, ADNet consists of two core components: a trainable despeckling network Ψ_{θ^*} (upper part) and a parameter-freeze white-box surrogate detection model $\mathcal{F}_\Theta$ (lower part). During the training phase, the original SAR image ($\mathbf{X}$) is first fed into the despeckling network to

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output the speckle-free image. Then, this image is passed to the surrogate detection model to generate the confidence loss $\mathcal{L}_{\text{conf}}$. By backpropagating $\mathcal{L}_{\text{conf}}$, the despeckling network transforms the benign despeckled image into an adversarial despeckled image $\Psi_{\theta^*}(\mathbf{X})$. Notably, the two cascaded networks are trained end-to-end, and the parameters of the surrogate detection model are frozen to prevent interference the despeckling network. During testing, the original SAR image is used as input. Adversarial despeckled images are directly generated based on the well-trained network, eliminating the need for time-consuming operations such as iterative optimization. To train the network Ψ_{θ^*}, we use two loss functions: the MAE-based $\mathcal{L}_{\text{des}}$ loss and $\mathcal{L}_{\text{conf}}$ loss, combined as follows:

$$\mathcal{L} = \mathcal{L}_{\text{des}} + \mathcal{L}_{\text{conf}}. \tag{3}$$

Similar to other speckle filtering methods [53], [54], [55], we use an MAE-based $\mathcal{L}_{\text{des}}$ loss to supervise the adversarial despeckling network, ensuring that the quality of the output SAR image is improved. For the original SAR image $\mathbf{X}$ and its ground-truth speckle-free SAR image $\mathbf{Y}$, let the despeckled image be $\tilde{\mathbf{X}} = \Psi_{\theta^*}(\mathbf{X})$. The loss function can be expressed as follows:

$$\mathcal{L}_{\text{des}}(\tilde{X}, \mathbf{Y}) = \frac{1}{H \times W} \sum_i^H \sum_j^W \left\| \tilde{X}_{i,j} - \mathbf{Y}_{i,j} \right\|_1. \tag{4}$$

The prediction vector of a deep object detector contains three learnable components, which describe the confidence of the target, the box coordinates, and the class probability. Each component can be regarded as a potential attack target. In this work, we add an adversarial loss $\mathcal{L}_{\text{conf}}$ to (4) to specifically attack the object confidence of the detector.

For each image in the training dataset, the deep target detector [13], [14], [15] first generates S prediction boxes. Subsequently, based on the center coordinates and the overlap rate with anchors, each labeled box in $\mathcal{O}$ is assigned the most suitable anchor and prediction vector. Here, $m_i = 1$ indicates that the ith prediction box corresponds to the labeled box; otherwise, it is 0. Thus, $\hat{\mathcal{O}} = \{\hat{o}_i | m_i = 1, 1 \le i \le S\}$ represents the set of matched prediction boxes, where $\hat{o}_i = (\hat{b}_i^x, \hat{b}_i^y, \hat{b}_i^h, \hat{b}_i^w, \hat{c}_i, \hat{k}_i)$ and $\hat{k}_i$ denotes the probabilities of the prediction boxes being true targets. Similar to prior works [36], the confidence loss $\mathcal{L}_{\text{conf}}$ is aimed at disrupting the confidence of the detector in positive sample boxes, rendering it ineffective, and is defined as follows:

$$\begin{aligned} \mathcal{L}_{\text{conf}}(\tilde{X}; \mathcal{F}_\Theta, \mathcal{O}) &= -\xi \frac{1}{N} \sum_{i=1}^S m_i \log\left[1 - \text{sig}(\hat{k}_i)\right] \\ &= -\xi \frac{1}{N} \sum_{i=1}^S m_i \cdot \log\left(1 - \frac{1}{1 + e^{-\hat{k}_i}}\right) \end{aligned} \tag{5}$$

where N denotes the number of bounding boxes and ξ denotes the weight assigned to loss, which is set empirically as 0.5.

C. Meta-Adversarial Despeckling Network

Using a surrogate (white-box) model to generate adversarial perturbations for transfer attacks on unknown (black-box) models is currently a primary method for black-box attacks [40], [43], [44]. However, significant differences in architectures and parameters between models greatly limit the success rate of adversarial examples in black-box attacks [40], [41], [42], [43], [44]. For the more complex object detection task, there is still considerable room for improvement in the cross-model attack performance of adversarial examples, as evidenced by the related experimental results in Section IV-D.

To address this issue, inspired by current work on meta-gradient attacks [42], we propose a meta-adversarial attack framework based on the baseline model ADNet to enhance the transferability of adversarial despeckled images. Meta-ADNet does not require iterative optimization for individual images and can be combined with any existing despeckling network to generate adversarial examples. As shown in Fig. 3, assume that the model set $\mathcal{F} = \{\mathcal{F}_{\Theta_1}, \mathcal{F}_{\Theta_2}, \ldots, \mathcal{F}_{\Theta_N}\}$ contains N surrogate detection models, where $N \ge 2$ and $N = 1$ represents the original ADNet. We randomly sample $n+1$ models iteratively to form T tasks. Each task includes two phases: metatraining and metatesting. Specifically, in the metatraining phase, the despeckled network is initially trained by ensemble attacks on n surrogate detection models. In the metatesting phase, the despeckling network is used to attack one unknown surrogate model further to simulate a black-box attack, making the adversarial despeckled images more transferable to other detection models.

1) Metatraining: Similar to the ensemble-based transfer attack approach, for the ith task, which includes K training samples, we use an ensemble of n surrogate (white-box) detection models $\mathcal{F}^i_{\Theta_1}, \mathcal{F}^i_{\Theta_2}, \ldots, \mathcal{F}^i_{\Theta_n}$ to guide the filtering network Ψ_{θ^*} in generating adversarial despeckling images. For the jth ($1 \le j \le K$) training sample $\mathbf{x}_{i,j}$, according to 5, the adversarial loss for the mth surrogate model $\mathcal{F}^i_{\Theta_m}$ can be represented as $\mathcal{L}_{\text{conf}}[\Psi_{\theta^*}(\mathbf{x}_{i,j}); \mathcal{F}^i_{\Theta_m}, \mathcal{O}_{i,j}]$. To combine the loss functions of all surrogate models, we fuse the adversarial losses as follows:

$$\mathcal{L}_{M_{i,j}}(\mathbf{x}_{i,j}) = \sum_{t=1}^n w_t \mathcal{L}_{\text{conf}}\left[\Psi_{\theta^*}(x_{i,j}); \mathcal{F}^i_{\Theta_t}, \mathcal{O}_{i,j}\right] \tag{6}$$

where $\Psi_{\theta^*}(\mathbf{x}_{i,j})$ represents the adversarial despeckling image generated by the network when the input is the original SAR image $\mathbf{x}_{i,j}$. The weight w_t corresponds to each surrogate model, with $w_t > 0$ and $\sum_{t=1}^n w_t = 1$. For all training examples $\mathbf{X}_i = \{\mathbf{x}_{i,1}, \mathbf{x}_{i,2}, \ldots \mathbf{x}_{i,K}\}$ in the metatraining process, we jointly optimize the despeckling network by minimizing both the confidence loss and the despeckling loss to ensure high transferability and visual quality of the despeckled images. The optimization objective for the metatraining process can be summarized as

$$\underset{\theta^*}{\text{argmin}}\, \mathbb{E}\left\{\mathcal{L}_{M_{i,1},\ldots,M_{i,K}} + \mathcal{L}_{\text{des}_{i,1},\ldots,\text{des}_{i,K}}\right\} \tag{7}$$

where $\mathcal{L}_{M_{i,1},\ldots,M_{i,K}} = \sum_{j=1}^K \mathcal{L}_{M_{i,j}}(x_{i,j})$ and $\mathcal{L}_{\text{des}_{i,1},\ldots,\text{des}_{i,K}} = \sum_{j=1}^K \mathcal{L}_{\text{des}}(x_{i,j})$. Substituting (4) and (6) into the above equation, it can be further expanded as follows:

$$\underset{\theta^*}{\text{argmin}}\, \mathbb{E}\{\sum_{j=1}^K \{\mathcal{L}_{M_{i,j}}(x_{i,j}) + \mathcal{L}_{\text{des}_{i,j}}\left[\Psi_{\theta^*}(x_{i,j}), y_{i,j}\right]\}\}$$

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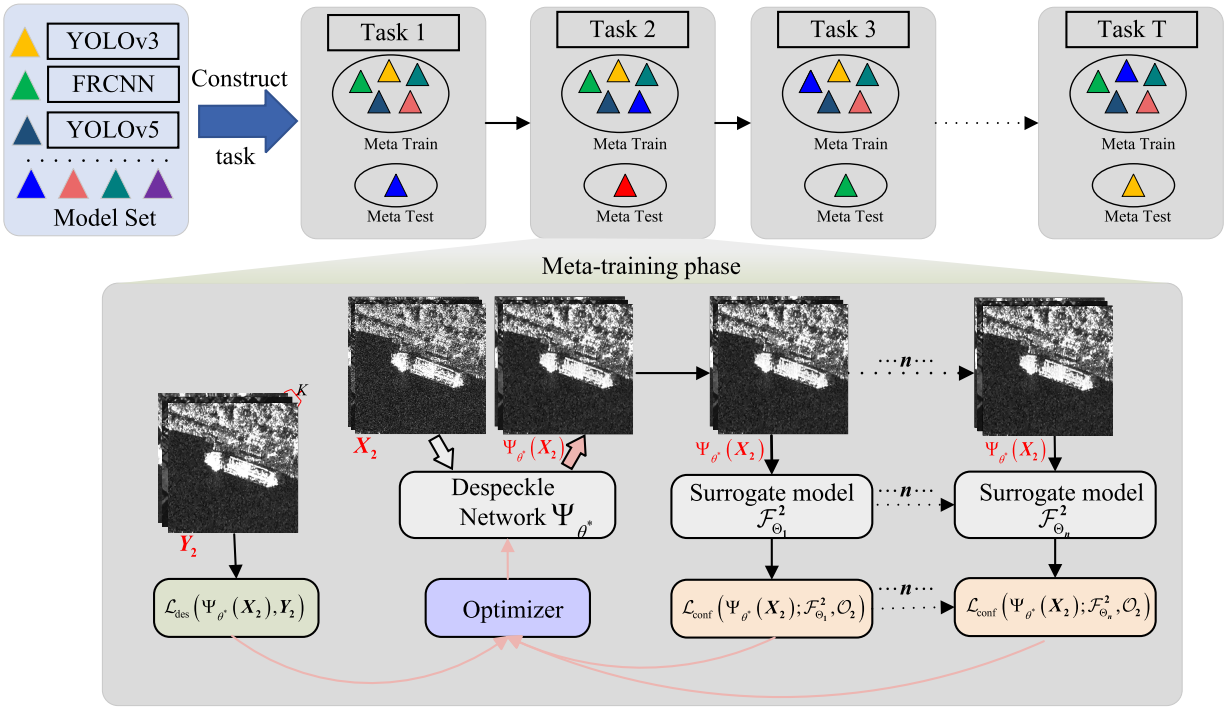


Fig. 3. Overview of the proposed Meta-ADNet. Meta-ADNet constructs T tasks by iteratively randomly selecting $n+1$ models. Each task includes two stages: metatraining and metatesting. In the metatraining stage, an adversarial despeckling network is initially trained by integrating attacks on n surrogate detection models. In the metatesting stage, the despeckling network is used to further attack one unknown surrogate model to simulate a black-box attack, making the generated adversarial images easier to transfer to other models.

$$= \left\{ \sum_{j=1}^{K} \sum_{t=1}^{n} w_t \mathcal{L}_{\text{conf}} \left[\Psi_{\theta^*}(x_{i,j}); \mathcal{F}^i_{\Theta_t}, \mathcal{O}_{i,j} \right] + \sum_{j=1}^{K} \mathcal{L}_{\text{des}} \left[\Psi_{\theta^*}(x_{i,j}), y_{i,j} \right] \right\} \tag{8}$$

2) Metatesting: After obtaining the preliminarily trained adversarial despeckling network through the ensemble attacks in the metatraining phase, we use the last sampled surrogate model $\mathcal{F}^i_{\Theta_{n+1}}$ to simulate black-box attacks, further enhancing the transferability of adversarial examples against unknown detection models. Specifically, the metatesting process involves an input test sample $\mathbf{x}_{i,K+1}$, where we further optimize the adversarial despeckling network by calculating the confidence loss of $\mathcal{F}^i_{\Theta_{n+1}}$

$$\mathcal{L}_{M_{i,K+1}}(\mathbf{x}_{i,K+1}) = \mathcal{L}_{\text{conf}} \left[\Psi_{\theta^*}(\mathbf{x}_{i,K+1}); \mathcal{F}^i_{\Theta_{n+1}}, \mathcal{O}_{i,K+1} \right]. \tag{9}$$

Analogous to the metatraining process, for the ith task, the objective optimization function for the metatesting process can be defined as

$$\underset{\theta^*}{\operatorname{argmin}} \mathbb{E} \left\{ \mathcal{L}_{M_{i,K+1}}(\mathbf{x}_{i,K+1}) + \mathcal{L}_{\text{des}} \left[\Psi_{\theta^*}(\mathbf{x}_{i,K+1}), \mathbf{y}_{i,K+1} \right] \right\}. \tag{10}$$

By iteratively optimizing the objective functions of metatraining and metatesting, the adversarial despeckling network is trained to obtain a specific weight. By loading this weight, the original benign despeckling network can output perturbed despeckled images to attack SAR image detectors of different architectures. The training procedure of Meta-ADNet is summarized in Algorithm 1.

Algorithm 1 Training of Meta-Adversarial Despeckling Network

Input: N surrogate models $\mathcal{F} = \{\mathcal{F}_{\Theta_1}, \mathcal{F}_{\Theta_2}, \ldots, \mathcal{F}_{\Theta_N}\}$, training set $\mathbb{D}$, optimizer Adam, number of training samples K, despeckle network Ψ_{θ^*};
Output: Optimized weight θ^*;
for $i \in \{1, \ldots T\}$ **do**
  Randomly select $n+1$ models $\mathcal{F}^i_{\Theta_1}, \mathcal{F}^i_{\Theta_2}, \ldots, \mathcal{F}^i_{\Theta_{n+1}}$ from the model set $\mathcal{F}_{\Theta_1}, \mathcal{F}_{\Theta_2}, \ldots, \mathcal{F}_{\Theta_N}$ to form a task, and randomly sample $K+1$ data $X_i = \{x_{i,1}, x_{i,2}, \ldots x_{i,K+1}\}$ from the training set $\mathbb{D}$ to form the training samples of the ith task
  for $j \in \{1, \ldots K\}$ **do**
    Calculate the confidence loss $\mathcal{L}_{M_{i,j}}$ and despeckling loss $\mathcal{L}_{\text{des}_{i,j}}$ of input $x_{i,j}$ on n white-box surrogate models $\mathcal{F}^i_{\Theta_1}, \mathcal{F}^i_{\Theta_2}, \ldots, \mathcal{F}^i_{\Theta_n}$ via Eqn 6 and Eqn 4
    Update network parameters $\theta^* \leftarrow \text{Adam}(\mathcal{L}_{M_{i,j}} + \mathcal{L}_{\text{des}_{i,j}}, \theta^*)$
  end for
  Calculate the confidence loss $\mathcal{L}_{M_{i,K+1}}$ and despeckling loss $\mathcal{L}_{\text{des}_{i,K+1}}$ of input $x_{i,K+1}$ on n+1th black-box surrogate models $\mathcal{F}^i_{\Theta_{n+1}}$ via Eqn 9 and Eqn 4
  Update network parameters via Eqn10 $\theta^* \leftarrow \text{Adam}(\mathcal{L}_{M_{i,K+1}} + \mathcal{L}_{\text{des}_{i,K+1}}, \theta^*)$
end for

3) Gradient Calculation: In this article, we demonstrate the feasibility of the proposed architecture through gradient differentiability. Taking the metatesting phase as an example,

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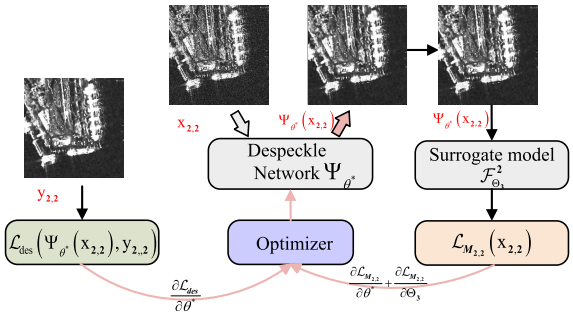


Fig. 4. Gradient schematic of Meta-ADNet during the metatesting phase for the second task.

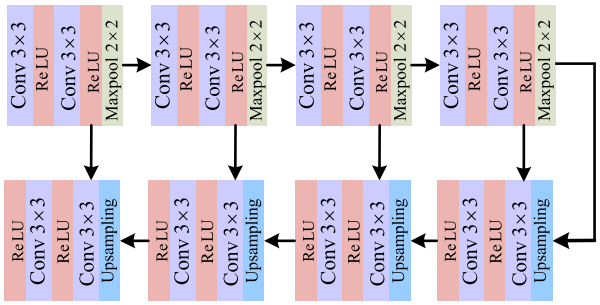


Fig. 5. Network architecture of U-Net.

the adversarial despeckling network's parameters are jointly optimized by minimizing both the confidence loss and the despeckling loss. As shown in Fig. 4, we assume that the second task ($i = 2$) to include three randomly sampled surrogate model ($n = 2$) and one training sample ($K = 1$) the gradient in (10) is given by $\nabla_{\theta^*}[\mathcal{L}_{M_{2,2}}(\mathbf{x}_{2,2}) + \mathcal{L}_{\text{des}}(\Psi_{\theta^*}(\mathbf{x}_{2,2}), \mathbf{y}_{2,2})]$, which can be further expanded as follows:

$$\frac{\partial \mathcal{L}_{\text{des}}}{\partial \theta^*} = \frac{\partial \mathcal{L}_{\text{des}}}{\partial \Psi} \cdot \frac{\partial \Psi}{\partial \theta^*} \tag{11}$$

$$\frac{\partial \mathcal{L}_{M_{2,2}}}{\partial \Theta_3} = \frac{\partial \mathcal{L}_{\text{conf}}}{\partial \Theta_3} = \frac{\partial \mathcal{L}_{\text{conf}}}{\partial \mathcal{F}^2} \cdot \frac{\partial \mathcal{F}^2}{\partial \Theta_3} \tag{12}$$

$$\frac{\partial \mathcal{L}_{M_{2,2}}}{\partial \theta^*} = \frac{\partial \mathcal{L}_{\text{conf}}}{\partial \theta^*} = \frac{\partial \mathcal{L}_{\text{conf}}}{\partial \mathcal{F}^2} \cdot \frac{\partial \mathcal{F}^2}{\partial \Psi} \cdot \frac{\partial \Psi}{\partial \theta^*}. \tag{13}$$

These there equations indicate that the confidence loss can be backpropagated with respect to both the despeckling network and the surrogate detection network parameters, allowing optimization via SGD-based optimizers. During training, we freeze the parameters of the surrogate detection network to prevent interference with the optimization of the despeckling network.

IV. Experiments

A. Dataset Description and Experimental Setup

1) Dataset Description: Given the widespread use and representativeness of the SSDD and HRSID datasets, we conducted experiments on these two SAR image ship detection datasets to evaluate the attack performance of the proposed method. The SSDD dataset, released by Li et al. [65], consists of images collected by RadarSat-2, TerraSAR-X, and Sentinel-1 sensors, covering HH, HV, VH, and VV polarization modes. It contains 1160 images and 2456 ship targets. The resolutions of these images range from 1 to 15 m, with widths of 214–668 pixels and heights of 160–526 pixels. In our experiments, all images were resized to 512 × 512 pixels, with 928 images used for training and 232 images for testing. The HRSID dataset, released by Wei et al. [66], originates from Sentinel-1, TerraSAR-X, and TanDEMX. It comprises 5604 image patches, each of size 800×800 pixels, cropped from larger images. Following the default partitioning criteria, 3642 images were used for training and 1962 images for testing. For the despeckling phase, considering the absence of corresponding clean SAR images for real SAR data, we followed the training methodology described in [53].

TABLE I

Models on SSDD and HRSID. The First Five White-Box Models Constitute the Model Set Used for Ensemble Attacks, With the Remaining Models Being Black-Box Models

	Model	Backbone	Model properties
white-box models	Faster-RCNN [12]	VGG16 [67]	two-stage
	YOLOv3 [13]	Darknet53 [13]	one-stage
	EfficientDet [16]	EfficientDet-D0 [16]	one-stage
	YOLOv4-tiny [14]	Darknet53-tiny [14]	one-stage
	YOLOv5 [15]	CSPDarknet53 [15]	one-stage
black-box models	Retinanet [17]	ResNet50 [68]	one-stage
	FCOS [18]	ResNet50 [68]	one-stage
	YOLOv7 [19]	CSPDarknet53-Plus [19]	one-stage

We synthesized 3665 labeled pairs using the speckle noise model from (1) and (2) to train the despeckling network.

2) Experimental Setup: In this study, all experiments are based on the PyTorch 1.12 framework and trained on an NVIDIA RTX A6000 using the Adam optimizer. Considering the effectiveness of the U-Net network in denoising tasks [69], it was employed as the backbone of the despeckling network, with a detailed structure shown in Fig. 5. ADNet was initially trained solely on the synthetic dataset using the $\mathcal{L}_{\text{des}}$ loss with a learning rate of 1e-4 for 100 epochs to train the despeckling network. The trained weights were then used in an unsupervised manner to obtain despeckled images of all real SAR images in the detection datasets, serving as reference label images. Subsequently, the entire framework was trained for 30 epochs using the confidence loss $\mathcal{L}_{\text{conf}}$ of the attack injection network, with a learning rate of 5e-5.

As for Meta-ADNet, we selected five different types of detectors {Faster-RCNN [12] (abbreviated as F-RCNN), YOLOv3 [13], EfficientDet, YOLOv4-tiny [14], YOLOv5 [15]} to form the model set. Three victim detectors, including RetinaNet [17] (two-stage), YOLOv7 [19] (single-stage), and FCOS [18] (anchor-free), were used to evaluate black-box attack performance. Unless otherwise specified, the number of iterations for task T is 30 epochs, and the training samples K and sampled model numbers n in the metatraining process are set to 3 and 4. All object detectors were trained on the despeckled images from SSDD and HRSID using a learning rate of 1e-4 for 200 epochs, employing data augmentation techniques such as horizontal and vertical flipping to enhance the robustness of the detectors. The confidence thresholds for the detection models used in black- and white-box attacks were set to 0.5 for fairness. Table I lists the specific information of the eight widely used single- and two-stage object detectors, covering both with and without anchors, as well as different multiscale fusion frameworks.

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TABLE II
WHITE-BOX ATTACK PERFORMANCE OF ADNET AND META-ADNET ON FIVE OBJECT DETECTORS. ADV DENOTES ADVERSARIAL DESPECKLED IMAGES GENERATED AGAINST THE VICTIM MODEL, AND M-ADV DENOTES META-ADVERSARIAL DESPECKLED IMAGES. LOWER VALUES OF AP(%) AND AP_{50}(%) INDICATE BETTER ATTACK PERFORMANCE

Attack Methods	Dataset	Victim Detector	AP(%)↓			AP_{50}(%)↓			ASR(%)↑		SSIM↑	
			Benign	Adv.	M-adv.	Benign	Adv.	M-adv.	Adv.	M-adv.	Adv.	M-adv.
FGSM	SSDD	F-RCNN	32.5	11.0	-	75.8	30.6	-	66.2	-	0.800	-
UAE	SSDD	F-RCNN	32.5	10.3	-	75.8	33.7	-	68.3	-	0.986	-
ADNet / Meta-ADNet	SSDD	F-RCNN	32.5	9.1	2.0	75.8	24.6	4.1	72.0	93.9	0.977	0.969
	SSDD	YOLOv3	50.9	7.1	7.3	93.8	19.5	19.6	86.1	85.7	0.976	0.969
	SSDD	EfficientDet	51.6	32.1	7.8	86.8	70.3	25.1	37.8	84.9	0.972	0.969
	SSDD	YOLOv4-tiny	50.2	8.4	4.1	91.2	27.6	16.2	83.3	91.8	0.973	0.969
	SSDD	YOLOv5	55.6	19.8	10.2	95.2	47.3	24.2	64.4	81.7	0.994	0.969
FGSM	HRSID	F-RCNN	23.4	15.2	-	53.7	37.4	-	35.1	-	0.780	-
UAE	HRSID	F-RCNN	23.4	20.3	-	53.7	40.0	-	13.3	-	0.996	-
ADNet / Meta-ADNet	HRSID	F-RCNN	23.4	16.2	2.3	53.7	40.0	5.2	59.5	90.2	0.987	0.981
	HRSID	YOLOv3	56.2	24.9	11.8	92.2	53.8	29.4	55.7	79.0	0.989	0.981
	HRSID	EfficientDet	53.6	37.8	20.6	84.3	70.5	45.6	29.5	61.6	0.984	0.981
	HRSID	YOLOv4-tiny	48.9	10.3	7.5	82.6	26.2	23.5	78.9	84.7	0.989	0.981
	HRSID	YOLOv5	56.1	15.7	14.9	87.1	34.2	34.1	72.0	73.4	0.987	0.981

B. Evaluation Parameters

We used AP, AP_{50}, precision, and recall as evaluation metrics for ship detection performance. The decrease in AP, referred to as attack success rate (ASR), was used as the performance metric for the attack algorithm. The peak signal-to-noise ratio (PSNR) and the structural similarity index (SSIM) were employed as metrics for evaluating image quality.

Precision represents the proportion of correctly detected targets among all targets predicted as positive. Recall represents the proportion of correctly detected targets among all ground-truth targets. Their definitions are given as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{14}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{15}$$

where true positives (TP), false negatives (FN), and false positives (FP) represent the number of correctly detected targets, missed targets, and falsely detected targets, respectively. The average precision (AP) represents the area under the precision–recall curve and is currently the most frequently used evaluation metric for target detection. Its definition is given as follows:

$$\mathrm{AP} = \int_0^1 P(R) dR. \tag{16}$$

Similar to the metrics defined on the COCO dataset, in this article, AP represents the average precision calculated at multiple IoU thresholds ranging from 0.5 to 0.95, in step size of 0.05. AP_{50} denotes the average precision at an IoU threshold of 0.5.

ASR represents the decrease in AP of the attacked detector, with a range from 0 to 1. The stronger the attack performance on adversarial despeckled images, the larger the value of ASR. It is defined as follows:

$$\mathrm{ASR} = 1 - \frac{\mathrm{AP}_{\mathrm{adv}}}{\mathrm{AP}_{\mathrm{ori}}} \tag{17}$$

where $\mathrm{AP}_{\mathrm{ori}}$ and $\mathrm{AP}_{\mathrm{adv}}$ represent the AP values of the detector on clean images and adversarial images, respectively.

PSNR evaluates the similarity between the generated adversarial despeckled image $\tilde{\mathbf{X}}$ and the label $\mathbf{Y}$

$$\mathrm{PSNR} = 20\log_{10}\left(\frac{\mathbf{Y}_{\max}}{\sqrt{\mathrm{MSE}}}\right) \tag{18}$$

where $\mathbf{Y}_{\max}$ represents the maximum power, which is 255. MSE stands for the mean square error of $\tilde{\mathbf{X}}$ and $\mathbf{Y}$, given as $\mathrm{MSE} = \mathrm{E}[(\tilde{\mathbf{X}} - \mathbf{Y})^2]$.

The SSIM measures the structural similarity between the adversarial despeckled image $\tilde{\mathbf{X}}$ and the label $\mathbf{Y}$, with a range of 0 to 1. It is defined as follows:

$$\mathrm{SSIM}(\tilde{\mathbf{X}}, \mathbf{Y}) = \frac{(2\mu_{\tilde{\mathbf{X}}}\mu_{\mathbf{Y}} + c_1)(2\sigma_{\tilde{\mathbf{X}}\mathbf{Y}} + c_2)}{(\mu_{\tilde{\mathbf{X}}}^2 + \mu_{\mathbf{Y}}^2 + c_1)(\sigma_{\tilde{\mathbf{X}}}^2 + \sigma_{\mathbf{Y}}^2 + c_2)} \tag{19}$$

where μ and σ represent the mean and variance of the image, respectively, and $\sigma_{\tilde{\mathbf{X}}\mathbf{Y}}$ represents the covariance between $\tilde{\mathbf{X}}$ and $\mathbf{Y}$. c_1 and c_2 are both positive constants.

C. Effectiveness of White-Box Attack

We conducted comparative experiments on the SSDD and HRSID test sets to assess the proposed methods' white-box attack capabilities. Specifically, ADNet and Meta-ADNet were employed to attack five different model architectures of white-box detection models, including the F-RCNN with Vgg16 backbone and four single-stage detectors: YOLOv3, YOLOv4-tiny, YOLOv5, and EfficientDet.

1) Quantitative Results: The image quality of the adversarial despeckled samples generated by the proposed methods and their white-box attack performance are shown in Table II. Based on the AP_{50} and SSIM metrics, ADNet can significantly reduce the performance of the detector while maintaining high image quality, especially against the YOLOv4-tiny, achieving an ASR of 83.3% (SSDD) and 78.94% (HRSID). Furthermore, with the addition of the proposed metatraining framework, the attack capability of adversarial samples based on Meta-ADNet is further enhanced. Taking EfficientDet

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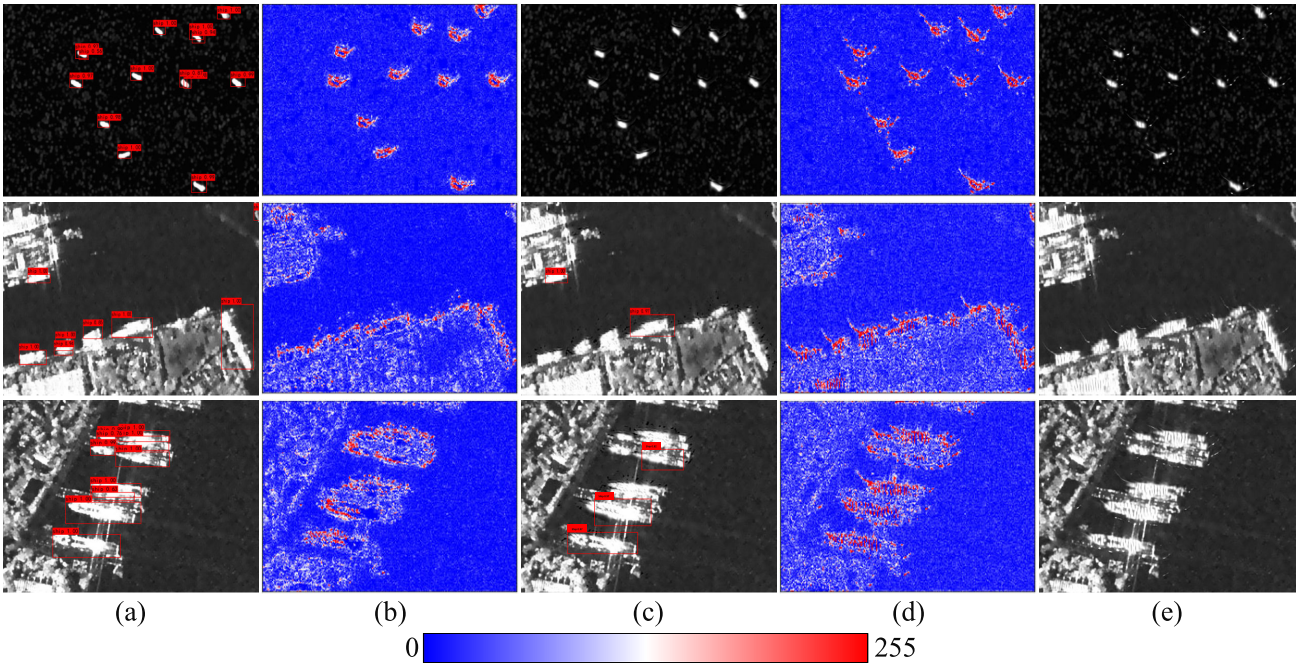


Fig. 6. Visual attack results of adversarial despeckled images on object detector (i.e., F-RCNN) in the SSDD dataset. (a) Detection results on the original despeckled SAR image. (b) Differences between the original despeckled SAR image and the adversarial despeckled SAR image generated by ADNet. (c) Detection results on the adversarial despeckled image generated by ADNet. (d) Differences between the original despeckled SAR image and the adversarial despeckled SAR image generated by Meta-ADNet. (e) Detection results on the adversarial despeckled image generated by Meta-ADNet. The color bar shows the range of differences in (b) and (d), which are amplified tenfold for visual clarity.

as an example, its AP_{50} decreases from 70.3% to 25.1% (86.8%→70.3%→25.1%, SSDD) and from 70.5% to 45.6% (84.3%→70.5%→46.5%, HRSID), indicating that our method also possesses strong attack capability against detection algorithms with multiscale dense feature fusion. To illustrate the effectiveness of ADNet and Meta-ADNet's attacks, we also compared two classic attack methods in Table II, including the gradient-based iterative attack algorithm FGSM [34] and the adversarial samples' generation method UAE based on generative models [60]. Considering that the comparative methods do not have the despeckling capability, we perform attack algorithms on despeckled SAR images to generate adversarial despeckled examples. Based on the victim model F-RCNN, the ASR metric shows that the image quality of the adversarial examples generated by our method is higher than FGSM with globally iterative perturbation, and the attack capability is stronger than UAE based on generative models, indicating that our method can produce covert and efficient adversarial despeckled examples.

2) Visualization and Analysis: Figs. 6 and 7 illustrate the detection results of the proposed methods on the SSDD and HRSID datasets, including three types of attack scenarios: small targets, near-shore complex interference, and densely arranged ships. When comparing the perturbation images in the first row of Figs. 6 and 7, it is evident that the proposed methods can accurately locate small targets and add perturbations to disable the detector. This capability is crucial for maintaining high image quality and strong attack capability. The middle row of Figs. 6 and 7 shows the performance of ADNet and Meta-ADNet under near-shore complex interference. It can be seen that the perturbations are mainly added in areas with high backscatter intensity of artificial targets. In contrast, fewer perturbations are added in areas with fewer texture details, such as farmland and rivers, making the adversarial despeckled images visually more similar to the original benign images. In addition, in the adversarial images, densely packed ship targets can also successfully deceive the detector even though their contour structures and scattering properties are easily distinguishable, as shown in the last row of Figs. 6 and 7. Overall, the quantitative results mentioned above are reflected in the visual results, validating the effectiveness of the proposed method in various complex scenarios.

D. Transferability of Black-Box Attacks

1) Quantitative Results: We further investigated the transfer attack capability of the despeckling images generated by ADNet and Meta-ADNet on black-box models. All detection models listed in Table I were selected as subjects for the study, and experiments were conducted on the SSDD and HRSID datasets. For ADNet, we attacked each detector separately and used the remaining seven detectors as black-box models. For Meta-ADNet, the detectors in the model set were used as white-box ensemble models, and the remaining three detectors were used as black-box victim models. ASR and AP_{50} were used to measure their transfer attack capability. The specific comparative results are shown in Table III.

As can be inferred from the results, Gaussian noise has little interference with the network's object detection capability. Without considering metatraining, the transfer attack capability of the adversarial despeckling images generated by

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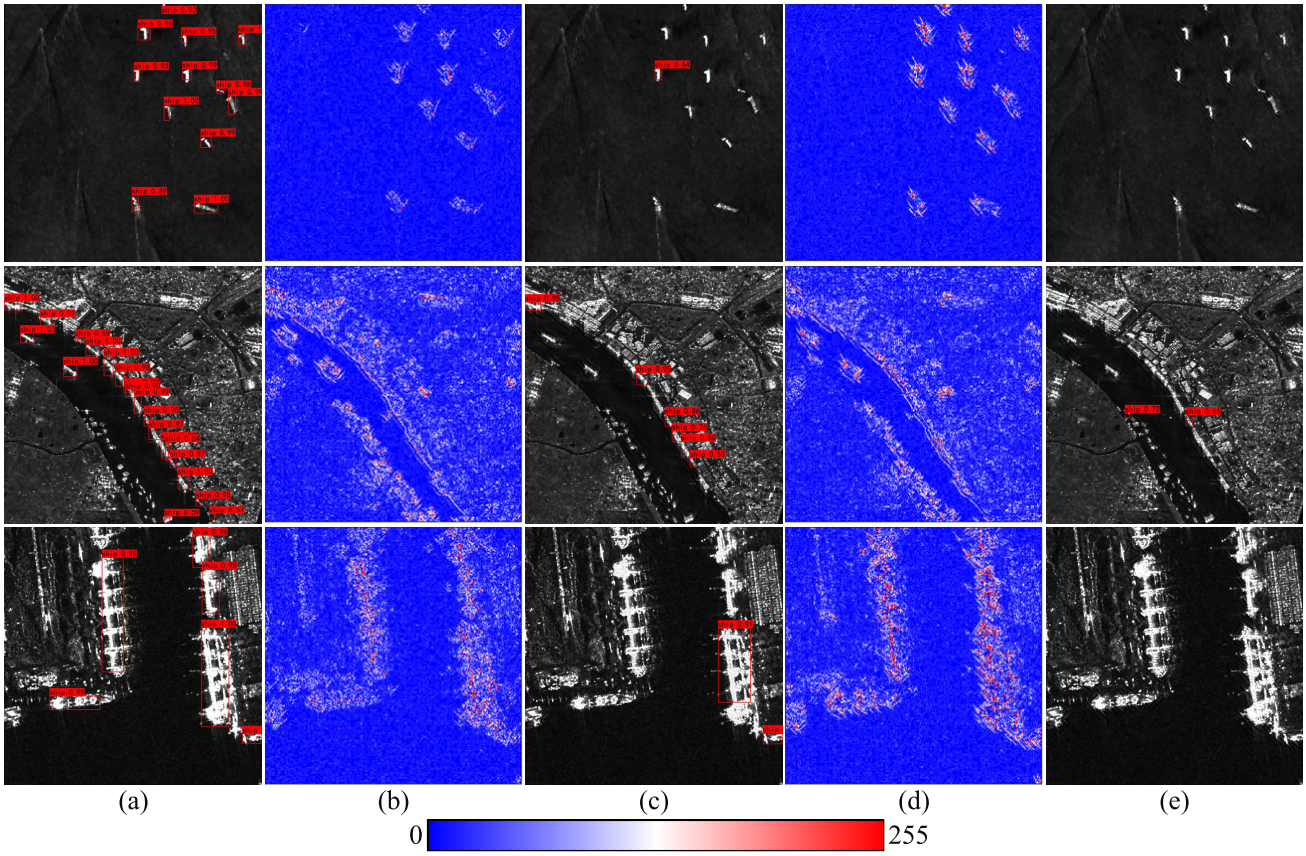


Fig. 7. Visual attack results of adversarial despeckled images on object detector in the HRSID dataset. The color bar shows the range of differences in (b) and (d), which are amplified tenfold for visual clarity. The definitions are the same as those in Fig 6. (a), (c), and (e) Detection results on the original despeckled image, the adversarial despeckled image generated by ADNet, and the adversarial despeckled image generated by Meta-ADNet, respectively.

TABLE III

TRANSFER ATTACK PERFORMANCE OF ADNET AND META-ADNET ON DIFFERENT OBJECT DETECTORS, THE VALUES TO THE LEFT-HAND SIDE OF / INDICATES AP_{50}(%) (THE LOWER, THE BETTER), AND THE VALUES TO THE RIGHT-HAND SIDE OF / INDICATES ASR(%) (THE HIGHER, THE BETTER). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. "A" IS AN ABBREVIATION FOR ADNET

Attack Methods	Dataset	Black-box models								Black-box Ave. ASR
		F-RCNN	YOLOv3	EfficientDet	YOLOv4-tiny	YOLOv5	Retinanet	FCOS	YOLOv7	
Random	SSDD	70.2/15.4	81.9/21.8	36.7/67.3	90.4/7.2	93.5/3.4	88.9/1.2	94.0/2.7	87.5/6.7	15.7
FGSM	SSDD	30.6/66.2	85.9/9.8	75.1/22.9	91.1/3.2	94.4/0.4	87.5/-0.4	95.0/1.3	91.3/3.2	5.8
UAE	SSDD	33.7/68.3	69.3/40.1	70.6/47.7	85.2/22.1	86.9/17.3	90.0/-14.6	93.0/5.3	90.5/1.2	17.0
A-F-RCNN	SSDD	24.6/72.0	84.8/23.2	77.5/31.8	87.7/14.9	91.5/14.6	76.3/27.7	92.1/15.7	89.4/7.8	19.4
A-YOLOv3	SSDD	49.3/42.8	19.5/86.1	80.2/17.8	82.7/22.7	86.3/18.0	83.2/11.3	94.5/5.5	87.9/13.0	18.7
A-EfficientDet	SSDD	38.4/49.2	84.1/22.2	70.3/37.8	85.9/20.1	88.1/16.6	72.8/30.4	92.6/13.9	87.0/15.1	23.9
A-YOLOv4-tiny	SSDD	20.6/72.6	39.6/69.4	61.9/56.2	27.6/83.3	56.7/96.0	70.2/44.2	70.9/47.0	46.9/69.1	64.9
A-YOLOv5	SSDD	22.7/70.5	75.8/34.6	55.1/60.1	59.9/63.9	47.3/64.4	70.2/46.2	85/31.7	24.0/88.2	56.4
Meta-ADNet	SSDD	**4.1/93.9**	**19.6/85.7**	**25.1/84.9**	**16.2/91.8**	**24.2/81.7**	**28.3/79.9**	**29.8/83.5**	**14.8/91.2**	**84.9**
Random	HRSID	52.0/6.0	88.4/7.1	77.5/12.1	80.1/3.5	83.4/5.4	73.0/22.9	81.1/7.0	74.9/9.8	9.2
FGSM	HRSID	37.4/35.0	81.5/19.0	70.0/21.5	76.8/10.4	79.3/11.9	63.7/39.4	72.8/18.2	68.9/20.3	22.0
UAE	HRSID	48.4/13.3	88.8/3.4	80.6/6.0	80.3/7.2	84.6/7.8	76.6/19.1	86.8/6.0	73.3/14.8	9.7
A-F-RCNN	HRSID	40.0/30.8	86.1/11.9	80.8/7.8	81.9/-0.4	82.3/8.0	70.1/25.5	85.3/6.2	76.8/2.9	8.8
A-YOLOv3	HRSID	38.1/38.5	53.8/55.7	76.5/18.7	81.2/0.8	77.1/16.8	67.1/29.4	83.8/10.9	76.1/5.1	17.2
A-EfficientDet	HRSID	38.8/28.2	67.4/43.1	70.5/29.5	77.6/13.1	74.6/24.2	57.6/46.5	79.3/20.2	67.1/24.8	28.6
A-YOLOv4-tiny	HRSID	26.2/56.0	56.4/48.8	66.4/34.5	23.2/83.0	60.4/40.3	4.5/96.2	43.2/61.5	65.0/26.8	52.1
A-YOLOv5	HRSID	21.7/66.7	49.2/58.5	68.2/42.2	77.3/18.0	34.2/73.4	14.1/88.3	63.0/44.0	74.2/8.7	46.6
Meta-ADNet	HRSID	**5.2/90.2**	**29.4/79.0**	**45.6/61.6**	**23.5/84.7**	**34.2/72.0**	**2.1/98.2**	**18.5/86.3**	**48.5/58.1**	**80.9**

ADNet on different surrogate models varies. Among them, the adversarial examples generated based on YOLOv4-tiny have the highest black-box average ASR of 75% on both datasets, surpassing the UAE comparative method at 40%. The ASRs on F-RCNN, YOLOv3, YOLOv5, and YOLOv7 reached 72.6%, 69.4%, 54.1%, and 69.1%, respectively, on the SSDD dataset. However, when facing detection networks with more considerable structural differences (such as FCOS and Retinanet), the attack effectiveness of adversarial examples based on ADNet is limited. In contrast, the transfer attack performance of the metalearning version Meta-ADNet is greatly improved, with the ASR metric far exceeding other methods. Particularly for

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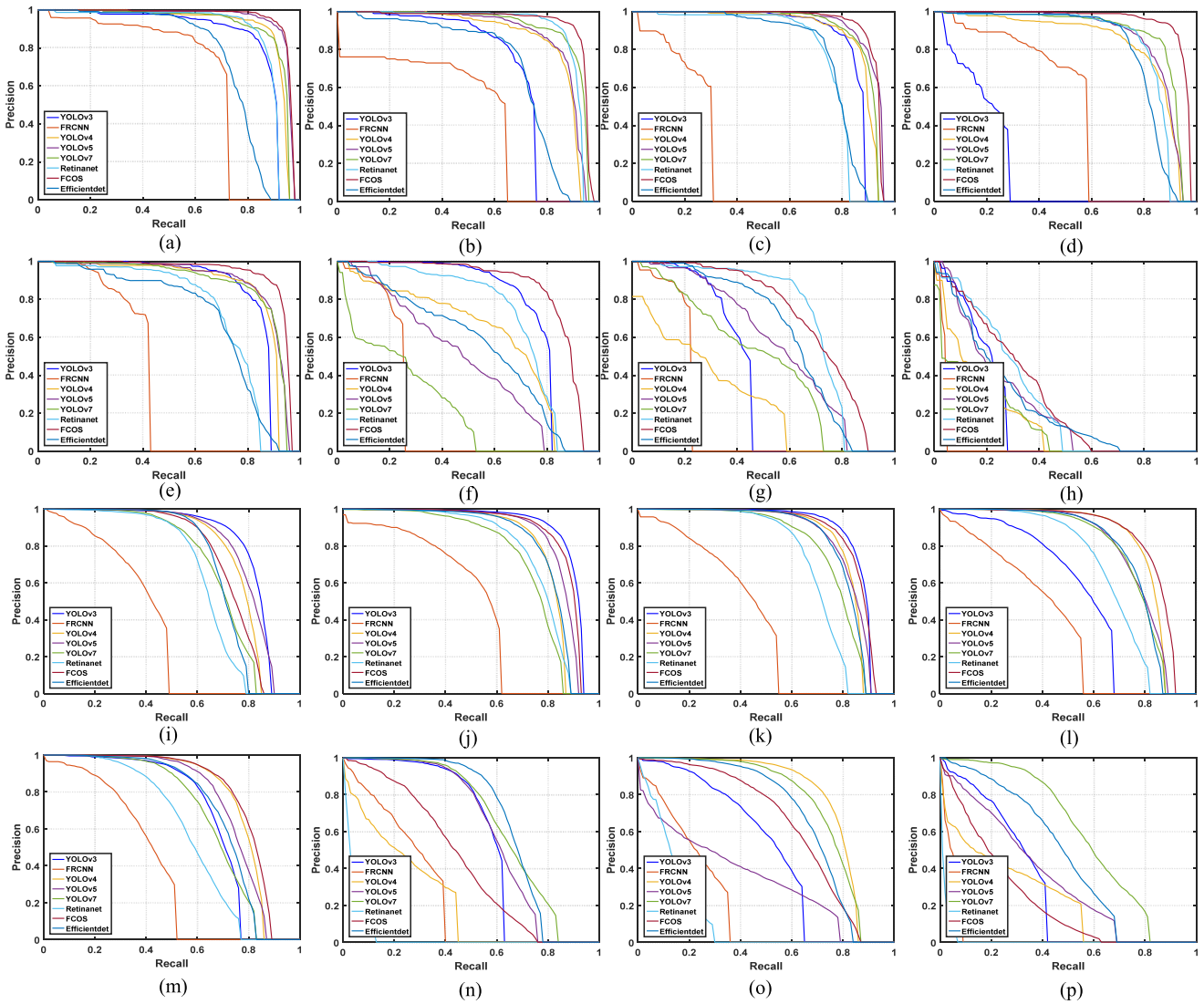


Fig. 8. PR curves of detection networks for different adversarial despeckled SAR images on SSDD and HRSID. Specifically, (a)–(h) and (i)–(p) represent the PR curves of adversarial examples generated by FGSM, UAE, ADNet-FRCNN, ADNet-YOLOv3, ADNet-EfficientDet, ADNet-YOLOv4-tiny, ADNet-YOLOv5, and Meta-ADNet on the SSDD and HRSID datasets, respectively.

the SSDD dataset, when attacking the anchor-free FCOS and the more complex model structure of YOLOv7, the ASR increased to 83.5% and 91.2%, respectively.

2) Visualization and Analysis: In Fig. 8, we illustrate the PR curves of the detection networks on different adversarial examples to intuitively demonstrate their attack performance. It can be observed that the adversarial despeckling examples generated by Meta-ADNet have the lowest detection accuracy at different recall rates, rapidly approaching zero. This indicates that the detection network has the most false background targets among all detected objects. Furthermore, we investigated the relationship between the efficiency frames per second (FPS) of different adversarial attack methods, the image quality of the generated adversarial images PSNR, and the black-box ASR. This relationship is presented in Fig. 9 as a bubble chart, where the larger bubble radius indicates higher operational efficiency, meaning a greater number of adversarial images generated per second. It can be seen that the proposed Meta-ADNet demonstrates significant advantages in both FPS and ASR metrics. In addition, the ADNet based on F-RCNN achieved higher ASRs and image quality compared to the UAE. In summary, the proposed method is an efficient and robust approach for generating adversarial examples that also incorporate despeckling.

E. Ablation Studies

To validate the importance of the despeckling loss $\mathcal{L}_{\text{des}}$ and the confidence loss $\mathcal{L}_{\text{conf}}$ for the proposed baseline network ADNet, we conducted ablation experiments using two widely used detectors, Faster-RCNN and YOLOv3, as white-box surrogate baseline models on the SSDD dataset and performed a sensitivity analysis on the weight coefficient ξ. The experimental results are shown in Table IV. "None" denotes the detection results on the original SAR images. When $\xi = 0$, only the despeckling loss $\mathcal{L}_{\text{des}}$ is included, AND ADNet w/o $\mathcal{L}_{\text{conf}}$ functions as a benign despeckling network, effectively removing speckles and clutter from SAR

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TABLE IV
ABLATION EXPERIMENTS OF LOSS FUNCTIONS IN ADNET

Model	Loss (ξ)	AP(%)↓	AP_{50}(%)↓	ASR(%)↑	PSNR↑
F-RCNN	None	29.2	72.1	-	-
	$\mathcal{L}_{\text{des}}$	32.5	75.8	0.0	-
	$\mathcal{L}_{\text{des}} + 0.5\mathcal{L}_{\text{conf}}$	9.1	24.6	72.1	44.2
	$\mathcal{L}_{\text{des}} + 1.0\mathcal{L}_{\text{conf}}$	6.7	17.5	79.4	42.6
	$\mathcal{L}_{\text{des}} + 1.5\mathcal{L}_{\text{conf}}$	5.8	16.1	82.2	41.5
	$\mathcal{L}_{\text{des}} + 2.0\mathcal{L}_{\text{conf}}$	4.3	11.8	86.8	41.3
YOLOv3	None	50.8	90.3	-	-
	$\mathcal{L}_{\text{des}}$	50.9	93.8	0.0	-
	$\mathcal{L}_{\text{des}} + 0.5\mathcal{L}_{\text{conf}}$	7.1	19.5	86.1	39.4
	$\mathcal{L}_{\text{des}} + 1.0\mathcal{L}_{\text{conf}}$	4.2	11.3	91.8	38.3
	$\mathcal{L}_{\text{des}} + 1.5\mathcal{L}_{\text{conf}}$	3.9	10.9	92.3	37.9
	$\mathcal{L}_{\text{des}} + 2.0\mathcal{L}_{\text{conf}}$	3.0	9.5	94.1	36.7

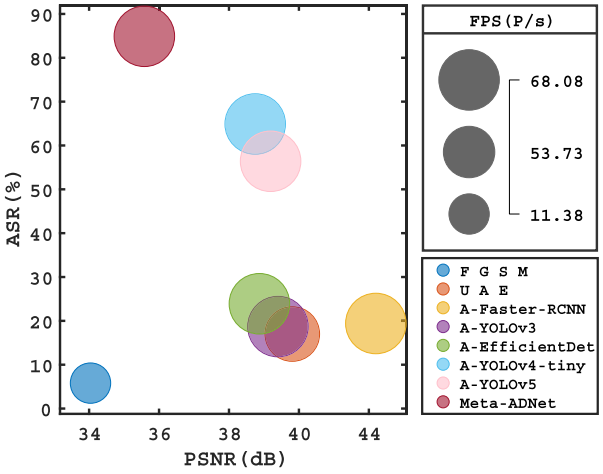


Fig. 9. Comparison of the proposed ADNet and Meta-ADNet with other adversarial attack methods. "A" is an abbreviation for ADNet.

images and improving the performance of the target detection methods. As seen, both F-RCNN and YOLOv3 achieve over a 3% improvement in AP_{50}. With the addition of the confidence loss $\mathcal{L}_{\text{conf}}$ ($\xi = 0.5$), ADNet generates adversarial despeckling samples that successfully attack the target detector. While maintaining a high image quality (PSNR of 44.21 and 39.43 Db, respectively), ASR reached 72% and 86.1%, verifying the effectiveness of $\mathcal{L}_{\text{conf}}$. As the value of ξ increases, the proportion of the confidence loss increases, leading to higher ASR, but the PSNR drops significantly. Taking the attack surrogate detection model YOLOv3 as an example, increasing ξ from 0.5 to 2 results in ASR rising from 86.05% to 94.11%, while PSNR drops from 39.43 to 36.67 dB. These quantitative results are visually reflected in Fig. 10. It is evident that as ξ increases, the number of streaks in the high-energy scattering regions also increases, resulting in poorer visual quality. Balancing these factors, we set ξ to 0.5 to ensure that the despeckling network produces adversarial despeckling images with both high-quality and high-attack capabilities.

In addition, we further analyzed and interpreted the reasons behind the failure of both detectors on adversarial images to understand the working principle of the confidence loss $\mathcal{L}_{\text{conf}}$. Fig. 11 illustrates the distribution of different detection types for F-RCNN and YOLOv3 on both the original despeckled

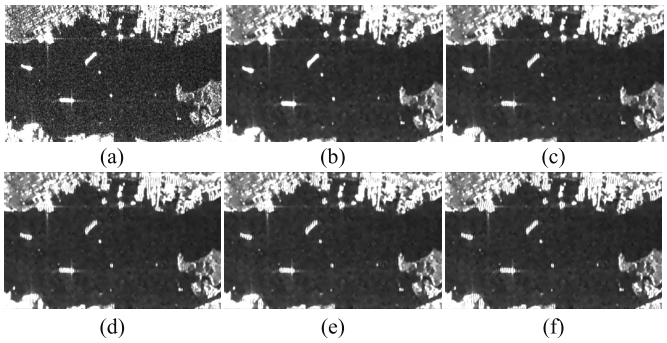


Fig. 10. Adversarial despeckling images generated under different values of ξ when attacking the surrogate detection model YOLOv3. (a) Original SAR image. (b) $\xi = 0$. (c) $\xi = 0.5$. (d) $\xi = 1.0$. (e) $\xi = 1.5$. (f) $\xi = 2.0$.

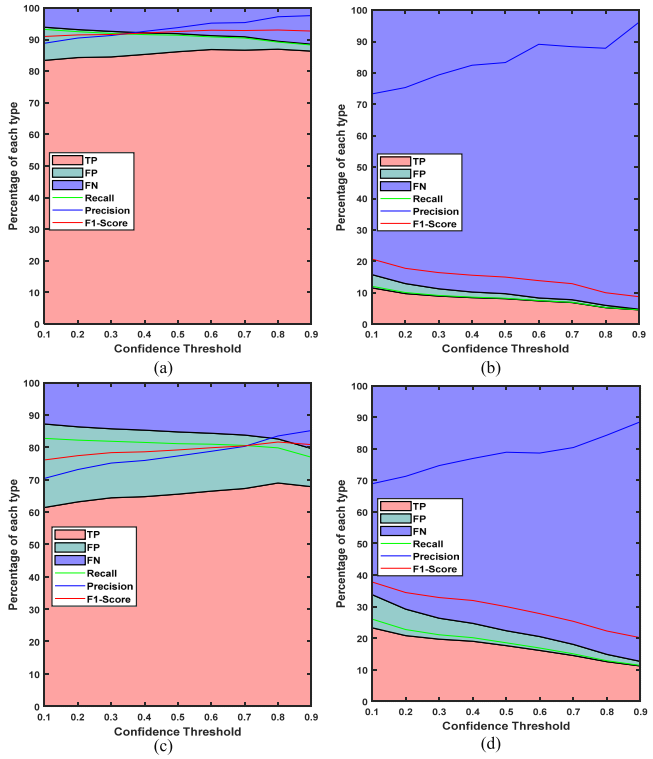


Fig. 11. Detection type distributions of the target detector on original images (first column) and adversarial images (last column) at different confidence thresholds. (a) and (b) YOLOv3. (c) and (d) F-RCNN.

TABLE V
ABLATION STUDY OF THE DESPECKLING BACKBONE NETWORK IN ADNET, WITH YOLOV3 AS THE SURROGATE DETECTION MODEL

Models	SSIM↑	PSNR ↑	ASR(%)↑
SAR-CNN [53]	0.986	38.31	71.5
MoNet [57]	0.984	38.43	70.7
SARDRN [56]	0.975	38.95	71.3

images and the adversarial despeckled images. For the original images, the detectors can accurately detect and identify ship targets even at high confidence thresholds (on the right-hand side of the image). However, for adversarial images, even when considering only low confidence thresholds, false negatives dominate, indicating that the detector tends to trigger in background areas unrelated to the target. This phenomenon

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TABLE VI

ASR(%) FOR BLACK-BOX VICTIM MODELS AND WHITE-BOX SURROGATE MODELS WITH DIFFERENT ENSEMBLE SIZES (N) ON SSDD. THE NUMBER OF SAMPLED TASKS T IS 30 EPOCHS. THE NUMBER OF TRAINING SAMPLES K IN THE METATRAINING STEP IS 3

n	White-box Models ASR(%)↑in Model Set					Black-box Models ASR(%)↑		
	F-RCNN	YOLOv3	EfficientDet	YOLOv4-tiny	YOLOv5	Retinanet	FCOS	YOLOv7
1	72.0	-	-	-	-	27.7	15.7	7.8
2	62.8	85.2	-	-	-	16.3	16.6	5.5
3	96.2	82.9	83.2	-	-	81.7	47.8	50.2
4	93.6	87.4	83.8	82.2	-	72.8	67.9	75.4
5	93.9	85.7	84.9	91.8	81.7	79.9	83.5	91.2

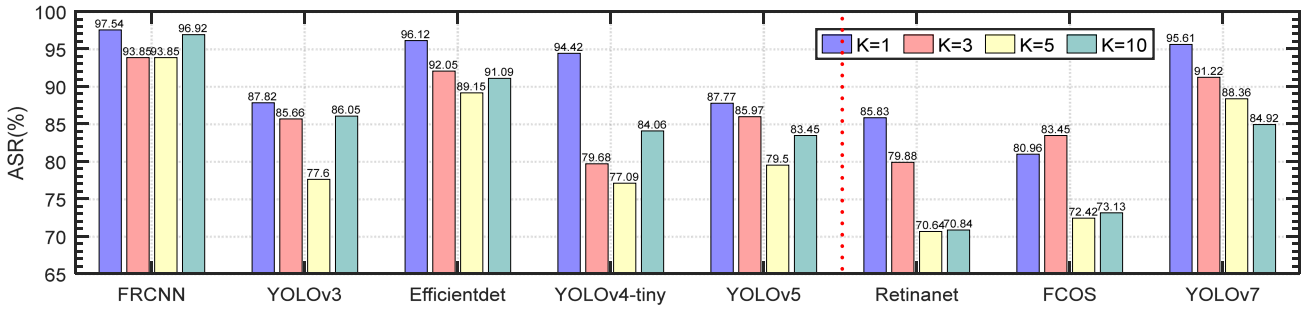


Fig. 12. ASR of white-box (left of the red dotted line) and black-box (right of the red dotted line) detection models with different numbers of training samples K during metatraining on SSDD.

indicates that the proposed method generates adversarial images by suppressing the confidence of the target to below the background and is robust to the threshold of the detectors.

Finally, we additionally incorporated three distinct despeckling backbones as baseline models to assess the robustness of the proposed ADNet algorithm. Specifically, we selected SAR-CNN [53], SARDRN [56], and MoNet [57] as despeckling networks while fixing the surrogate detection model for ablation experiments on the SSDD dataset. The experimental results are presented in Table V. As shown, for different despeckling networks, the adversarial despeckling samples generated by ADNet preserve high image quality while yielding an ASR exceeding 70%, providing quantitative validation of the proposed algorithm's applicability to various despeckling network architectures.

F. Impact of Hyperparameters on Meta-ADNet

The metatraining process of Meta-ADNet involves two crucial hyperparameters: the number of ensemble models n and the number of training samples K. In this section, we provide a detailed analysis of these hyperparameters' effects on Meta-ADNet's performance, offering guidance on their optimal selection.

1) Number of Ensemble Models n in Metatraining: In the metatraining phase, we utilize the ensemble loss from n surrogate models to train the adversarial despeckling network, producing adversarial examples. Table VI compares the average ASR of adversarial despeckling samples on white- and black-box detection models for different values of n. It is evident that the ASR on both white- and black-box models is directly proportional to the number of ensemble models n. The larger n is, the higher the success rate. This observation is consistent with the concept of ensemble attacks: the more models the adversarial samples can attack, the higher the likelihood of successfully transferring the attack to other black-box models.

TABLE VII

AVERAGE ASR(%) OF THE WHITE- AND BLACK-BOX MODELS CORRESPONDING TO DIFFERENT TRAINING SAMPLES K DURING THE METATRAINING STEP ON SSDD. THE NUMBER OF SAMPLED TASKS T IS 30 EPOCHS. THE NUMBER OF SAMPLED MODELS n IS 5

K	SSIM↑	PSNR↑	ASR(%)↑
$K = 1$	0.961	35.4	87.5
$K = 3$	0.969	35.6	84.9
$K = 5$	0.967	36.7	77.2
$K = 10$	0.968	36.1	76.3

2) Number of Training Samples K in Metatraining: During the metatraining phase, the number of training samples K represents the number of iterations for the ensemble models to optimize the despeckling network. Table VII compares the average ASR of adversarial despeckling samples on white- and black-box detection models for different iterations K, with detailed ASR metrics shown in Fig. 12. Interestingly, increasing the number of training samples K does not enhance the adversarial samples' attack performance; rather, it slightly decreases. We hypothesize that smaller K values result in more frequent alternation between metatraining and metatesting steps for the adversarial despeckling network, leading to stronger transferability of the generated perturbations. From a practical standpoint, we recommend setting $K = 3$, which provides the highest PSNR value, ensuring optimal image quality while maintaining effective adversarial attack performance.

V. CONCLUSION

In this article, speckle filtering is introduced into the adversarial attack on SAR image target detection, and a novel

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network framework called Meta-ADNet is proposed to generate adversarial despeckling images. Meta-ADNet consists of the baseline model ADNet and a meta-adversarial training framework. ADNet uses a cascaded approach that combines the despeckling network with the surrogate detection network, utilizing confidence loss to disrupt target prediction. The meta-adversarial training framework constructs multiple tasks to simulate white- and black-box attacks, training the despeckling network to enhance the transferability of adversarial examples. Quantitative evaluation results on two authoritative SAR image ship detection datasets, SSDD and HRSID, demonstrate that the proposed method generates adversarial despeckling images with strong attack transferability, capable of quickly and effectively attacking various advanced target detection algorithms. This provides an effective approach for evaluating the robustness of SAR image target detection algorithms. In future work, we will focus on understanding the origins of vulnerabilities in deep networks and developing more secure models.

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